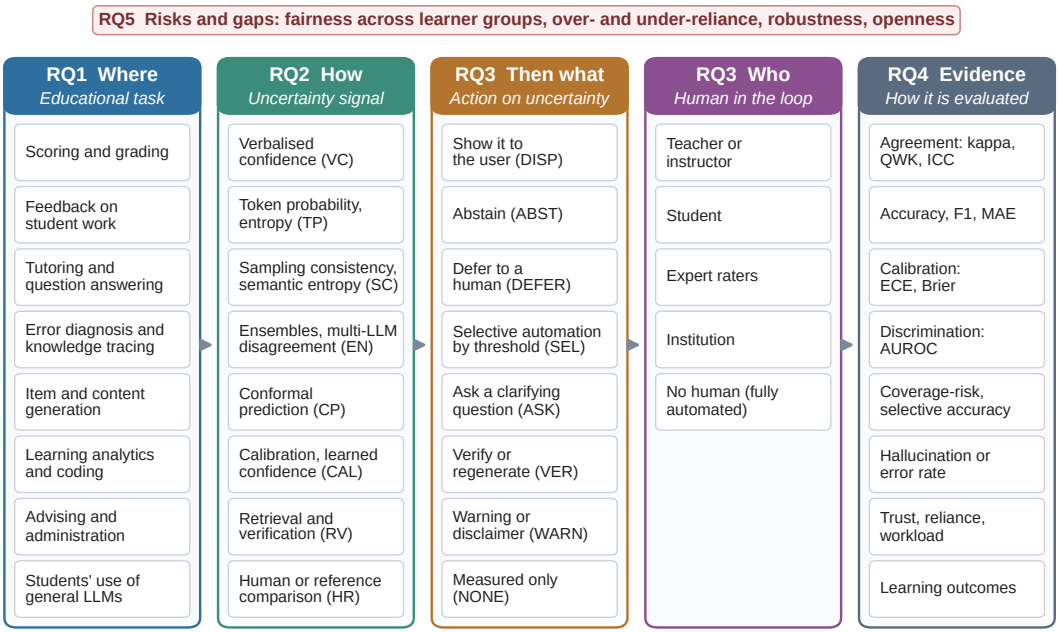
RQ5 Risks and gaps: fairness across learner groups, over- and under-reliance, robustness, openness
RQ1 Where
Educational task
Scoring and grading
Feedback on student work
Tutoring and question answering
Error diagnosis and knowledge tracing
Item and content generation
Learning analytics and coding
Advising and administration
Students' use of general LLMs
RQ2 How
Uncertainty signal
Verbalised confidence (VC)
Token probability, entropy (TP)
Sampling consistency, semantic entropy (SC)
Ensembles, multi-LLM disagreement (EN)
Conformal prediction (CP)
Calibration, learned confidence (CAL)
Retrieval and verification (RV)
Human or reference comparison (HR)
RQ3 Then what
Action on uncertainty
Show it to the user (DISP)
Abstain (ABST)
Defer to a human (DEFER)
Selective automation by threshold (SEL)
Ask a clarifying question (ASK)
Verify or regenerate (VER)
Warning or disclaimer (WARN)
Measured only (NONE)
RQ3 Who
Human in the loop
Teacher or instructor
Student
Expert raters
Institution
No human (fully automated)
RQ4 Evidence
How it is evaluated
Agreement: kappa, QWK, ICC
Accuracy, F1, MAE
Calibration: ECE, Brier
Discrimination: AUROC
Coverage-risk, selective accuracy
Hallucination or error rate
Trust, reliance, workload
Learning outcomes